WorkFlow

Field Team Coordination App

Project Documentation

Adam Strelec Damián Styrak

February 2026

Contents

1 Application Overview **2**

2 User Roles **2**
2.1 Manager 2
2.2 Worker 2

3 Screen Design **3**
3.1 Screen 1 — Dashboard (Manager) 3
3.2 Screen 2 — Project List 3
3.3 Screen 3 — Project Detail 3
3.4 Screen 4 — Project Creation 4
3.5 Screen 5 — Task List 4
3.6 Screen 6 — Task Creation 4
3.7 Screen 7 — Assign Worker 4
3.8 Screen 8 — Map 5
3.9 Screen 9 — Chat 5
3.10 Screen 10 — Worker List 5
3.11 Screen 11 — Worker Profile (Manager view) 5
3.12 Screen 12 — Profile (Worker view) 5
3.13 Screen 13 — Certificates (Worker — Send) 6
3.14 Screen 14 — Certificates (Manager — Review) 6

4 Prototype **6**

5 Acceptance Tests (UAT) **12**

Application Overview

WorkFlow is a cross-platform mobile application (iOS / Android) designed for coordinating field work teams. It replaces inefficient phone- or email-based coordination by giving managers an instant, real-time overview of task progress and team locations.

Core features:

- Creating, editing, and assigning projects and tasks within them to field workers.
- Live tracking of worker GPS positions on an interactive map.
- Task status updates and photo documentation upload directly from the field.
- Built-in real-time chat (WebSocket) between managers and workers.
- Push notifications for new assignments and status changes.
- Offline mode — changes are saved locally and synced automatically on reconnect.

Technology Stack

Layer	Technology
Frontend	React Native (iOS / Android)
Backend	Node.js (Express)
Database	PostgreSQL
Push notifications	FCM (Android) + APNs via FCM (iOS)
Authentication	Google SSO (OAuth 2.0)
Maps	react-native-maps (Google Maps / Apple Maps)

User Roles

The application defines two user roles with distinct permissions.

Manager

The Manager has full control over projects, tasks, and the team:

- Creates, edits, and deletes projects.
- Creates, edits, and deletes tasks within projects (title, description, GPS coordinates, deadline).
- Assigns tasks to one or more workers.
- Views live worker positions and task-site pins on the map.
- Has an overview of all tasks and their current statuses.
- Reviews and accepts worker skill certificates.
- Communicates with workers via the built-in chat.
- Receives a push notification when a worker marks a task as *Done*.

Worker

The Worker has restricted access — only to tasks assigned to them:

- Views the list of their assigned tasks.
- Opens task details: description, map, photos, current status.
- Changes the task status to *In Progress* or *Done*.

- Uploads photo documentation from the camera or gallery.
- Uploads skill certificates for manager review.
- Shares GPS location with the manager (every 10 seconds).
- Communicates with the manager via chat.
- Receives a push notification when a new task is assigned.
- Can update task status offline; the change syncs after reconnection.

Screen Design

The application contains the following functional screens (splash and login screens are not counted). Screens differ in available actions depending on the user role.

Screen 1 — Dashboard (Manager)

Description: The home screen for the Manager after login. Provides a quick summary of the current state of all work.

UI elements:

- Manager name, role badge, and online status.
- Summary card showing active vs. total tasks (e.g. 4/13 active, 9/13 pending) with a **Detail** button.
- Summary card showing total number of workers with a **Detail** button.
- Bottom navigation bar: Profile, Home, Chat.

Screen 2 — Project List

Description: List of all projects. Manager sees an **Add project** + button; workers see only projects they are assigned to.

UI elements:

- Scrollable list of projects, each with name and navigation arrow.
- **Add project** + button in the top right (Manager only).
- Tap a project — navigates to Project Detail.

Screen 3 — Project Detail

Description: Full detail view of a project. The layout differs by role.

UI elements (Manager):

- Project title, status, deadline, location, created by, created at.
- Client info: name, phone, email, construction site address.
- Progress section: tasks done counter (e.g. 3/7) and number of assigned workers.
- List of assigned workers with online/offline status indicators.
- Photo documentation gallery with + button to add photos.
- Action buttons: **Edit project**, **Delete project**, **Assign worker**, **Add task**, **View map**.

UI elements (Worker):

- Project title, status, deadline, location, created by.

- Client name and phone.
- Progress bar showing percentage of tasks completed.
- Tasks done counter with navigation to task list.
- Photo documentation gallery with + button.

Screen 4 — Project Creation

Description: Form for creating a new project. Available to Manager only.

UI elements:

- Project info: Title, Description, Deadline, Status (radio button).
- Location: Construction site address, GPS Latitude, GPS Longitude.
- Client info: Client name, Phone, Email.
- **Save** button.

Screen 5 — Task List

Description: List of tasks within a project. Accessible from Project Detail.

UI elements:

- Project name in the header.
- Filter icon and + button (Manager only) to add a new task.
- Expandable task cards showing: title, status indicator, category, description, assigned person, deadline, photo thumbnails, **Upload more photos** link, and **Mark as done** button.
- Collapsed tasks show only title and status.

Screen 6 — Task Creation

Description: Form for creating a new task within a project. Manager only.

UI elements:

- Title and Description fields.
- Category dropdown (e.g. Elektrika, Kurenie) and Deadline picker.
- Priority selector: Low / Medium / High.
- Assign worker section — shows available workers with their verified skill certificates.
- **Save Task** button.

Screen 7 — Assign Worker

Description: Screen for assigning a worker to a task. Manager only.

UI elements:

- Search bar for filtering workers.
- Skill filter chips (e.g. All, Elektrika, Voda, Kurenie, Plyn).
- *Available workers* section — each card shows name and verified skill badges; selectable via checkbox.
- *Missing certificate — not eligible* section showing workers who cannot be assigned due to missing qualifications.

- **Done** button to confirm assignment.

Screen 8 — Map

Description: Project-specific map showing live worker positions. Manager only.

UI elements:

- Project name in the header (e.g. *Project1 map*).
- Full-screen map view with worker markers.
- *Workers on this project* section below the map listing each worker with name and online/offline status.

Screen 9 — Chat

Description: Direct real-time messaging between Manager and Worker.

UI elements:

- Header with recipient's name, avatar, and online status.
- Message history in chronological order (alternating left/right alignment).
- Chat input field and **Send** button.

Screen 10 — Worker List

Description: Manager's overview of all their workers.

UI elements:

- Expandable worker cards — expanded view shows verified and missing skill certificates (e.g. Voda ✓, Kurenie ×).
- **Viac** (More) link for additional certificates.
- **Invite new worker** button at the bottom — generates a unique invite link (e.g. `workflow://invite?token=...`) that can be copied and shared.

Screen 11 — Worker Profile (Manager view)

Description: Manager's view of a worker's profile.

UI elements:

- Profile photo, first name, last name, hourly rate.
- List of skill categories with certificate status (verified / not verified) and **Zobraziť** (Show) link per certificate.

Screen 12 — Profile (Worker view)

Description: Worker's own profile and settings.

UI elements:

- Profile photo, username, last name, date of birth, hourly rate, bank information.
- **Edit profile** button.
- List of skill categories with certificate status.
- **Log out** button.

Screen 13 — Certificates (Worker — Send)

Description: Screen for uploading a skill certificate for manager review.

UI elements:

- Informative text about the certificate request.
- Sender name label.
- Large photo upload area with + button (camera or gallery).

Screen 14 — Certificates (Manager — Review)

Description: Screen for reviewing a certificate submitted by a worker.

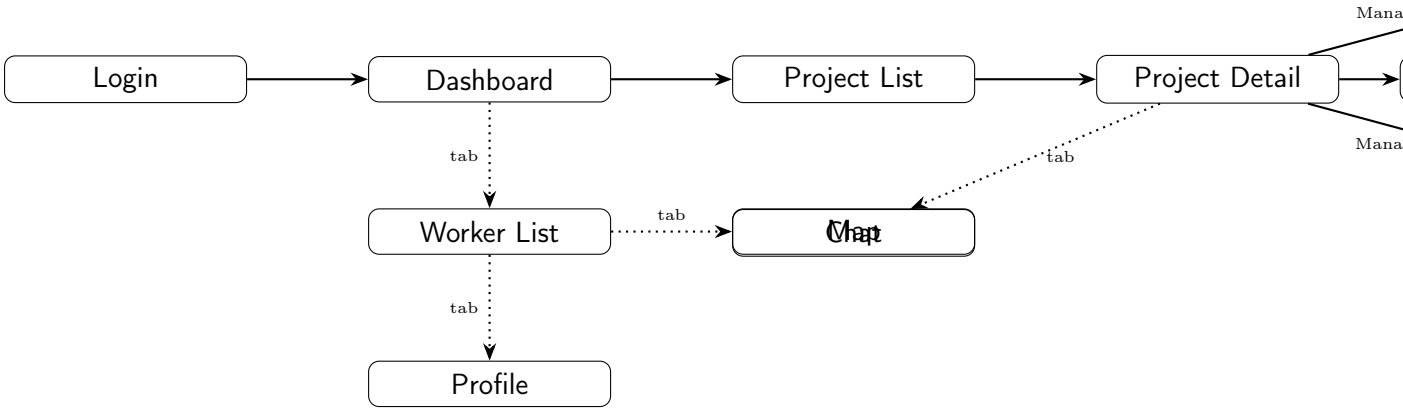
UI elements:

- Informative text and sender name.
- Certificate photo preview.
- **Accept** (✓) and **Reject** (×) action buttons.

Prototype

Navigation Flow

The app uses a **bottom tab bar** (Profile, Home, Chat) for top-level navigation and **stack navigation** within each section.



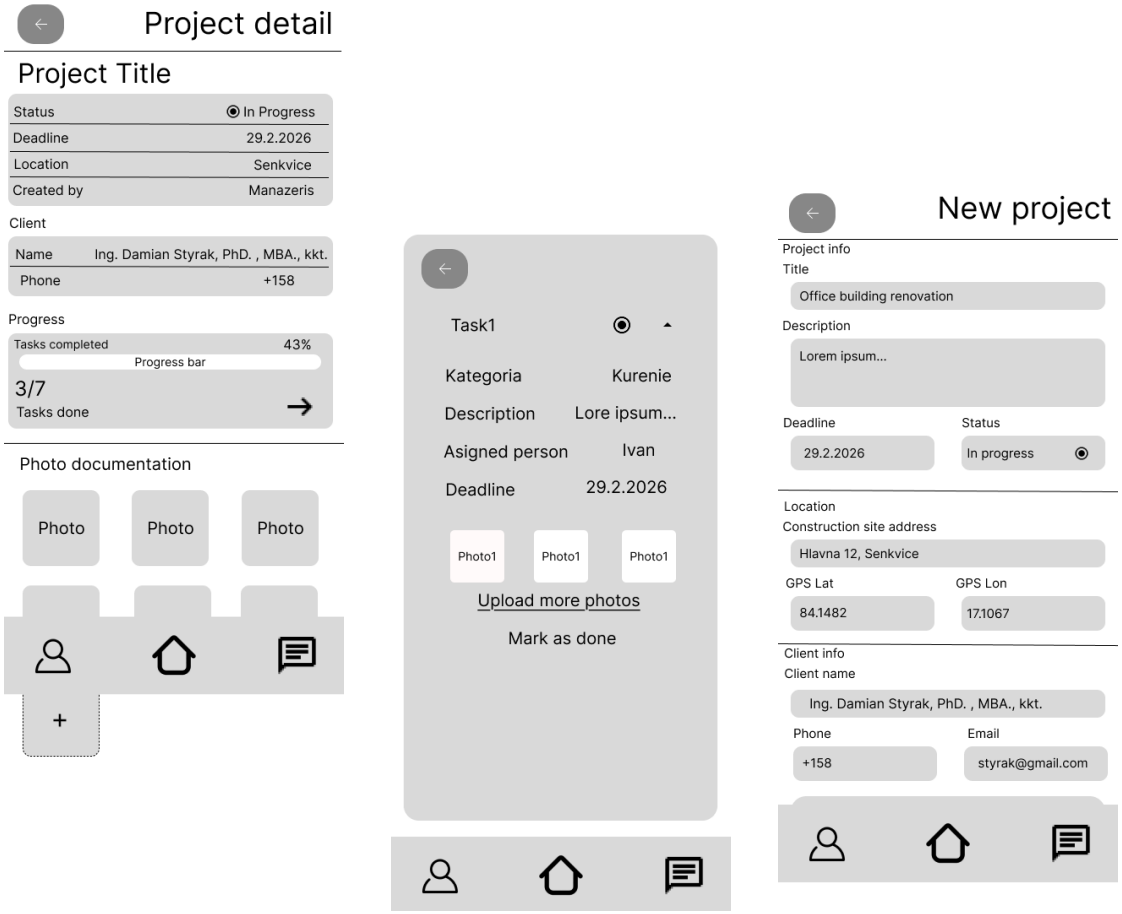
Prototype Screens

The following screens represent the actual designed prototype of the WorkFlow application.

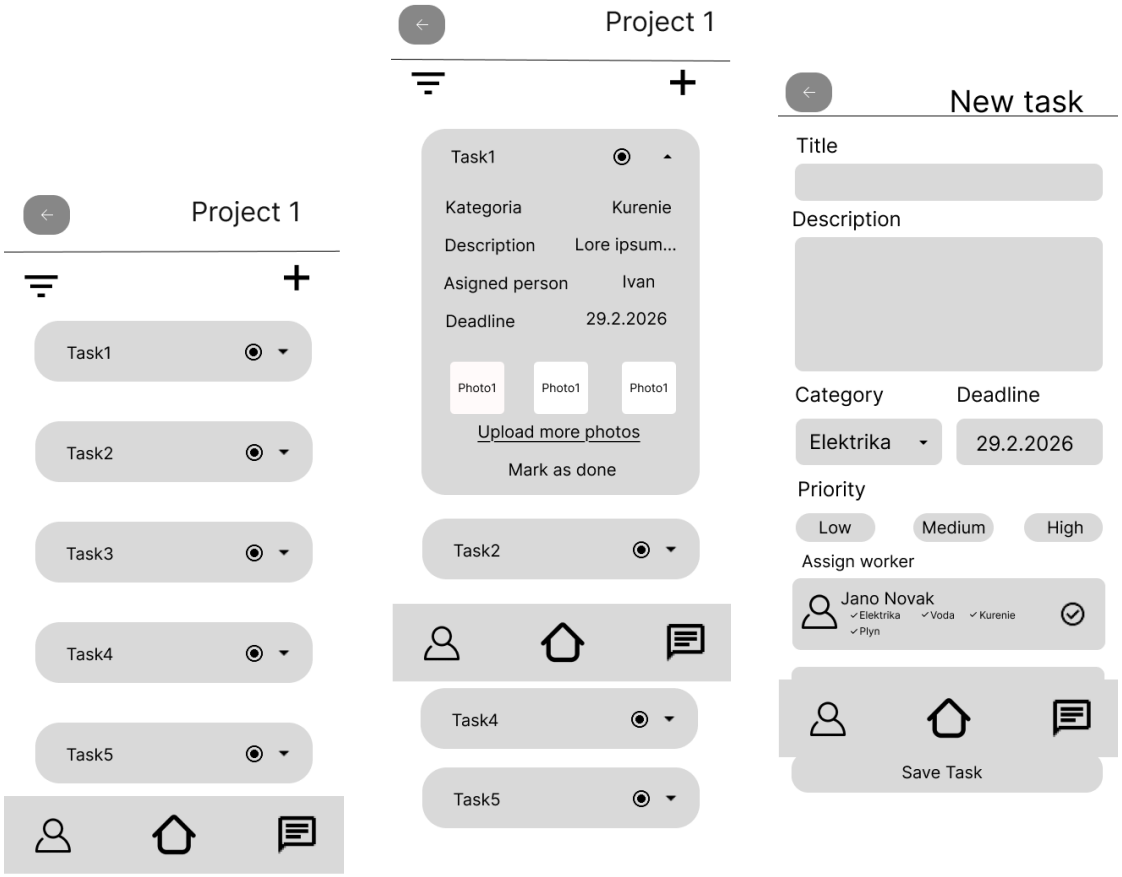
Dashboard & Project List



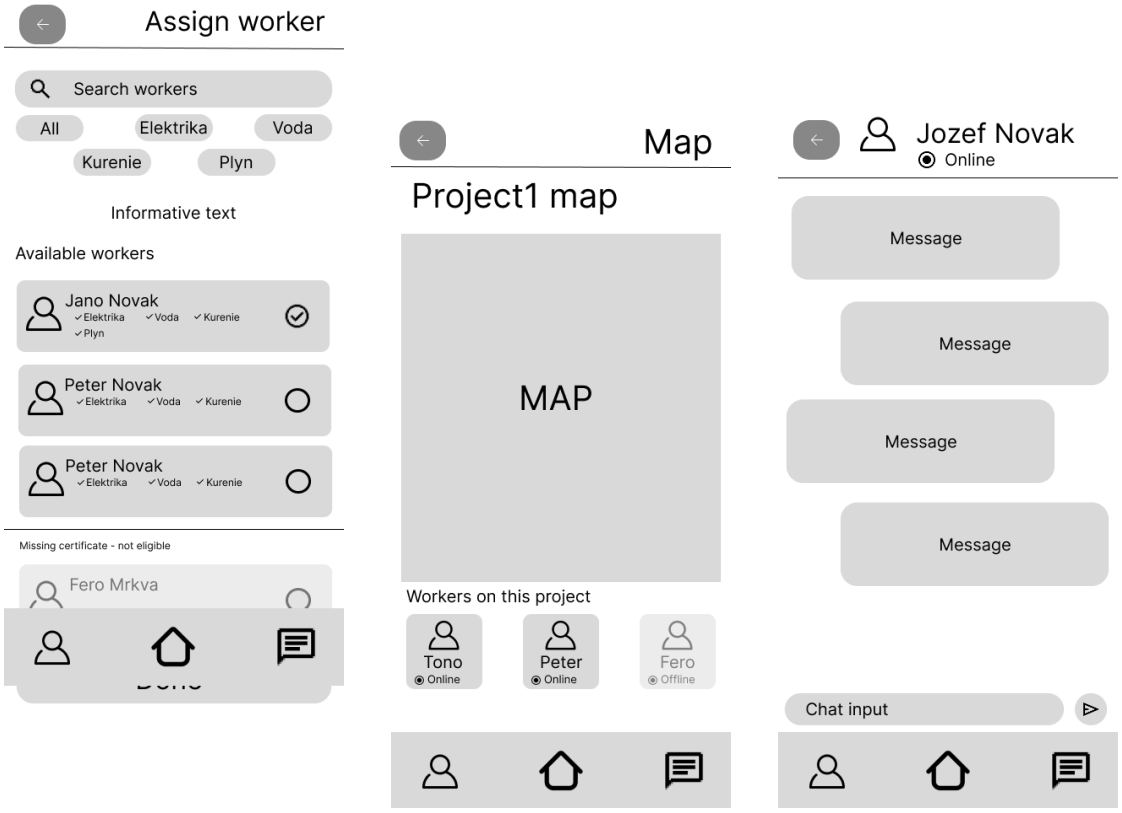
Project Detail (Manager & Worker) and Project Creation



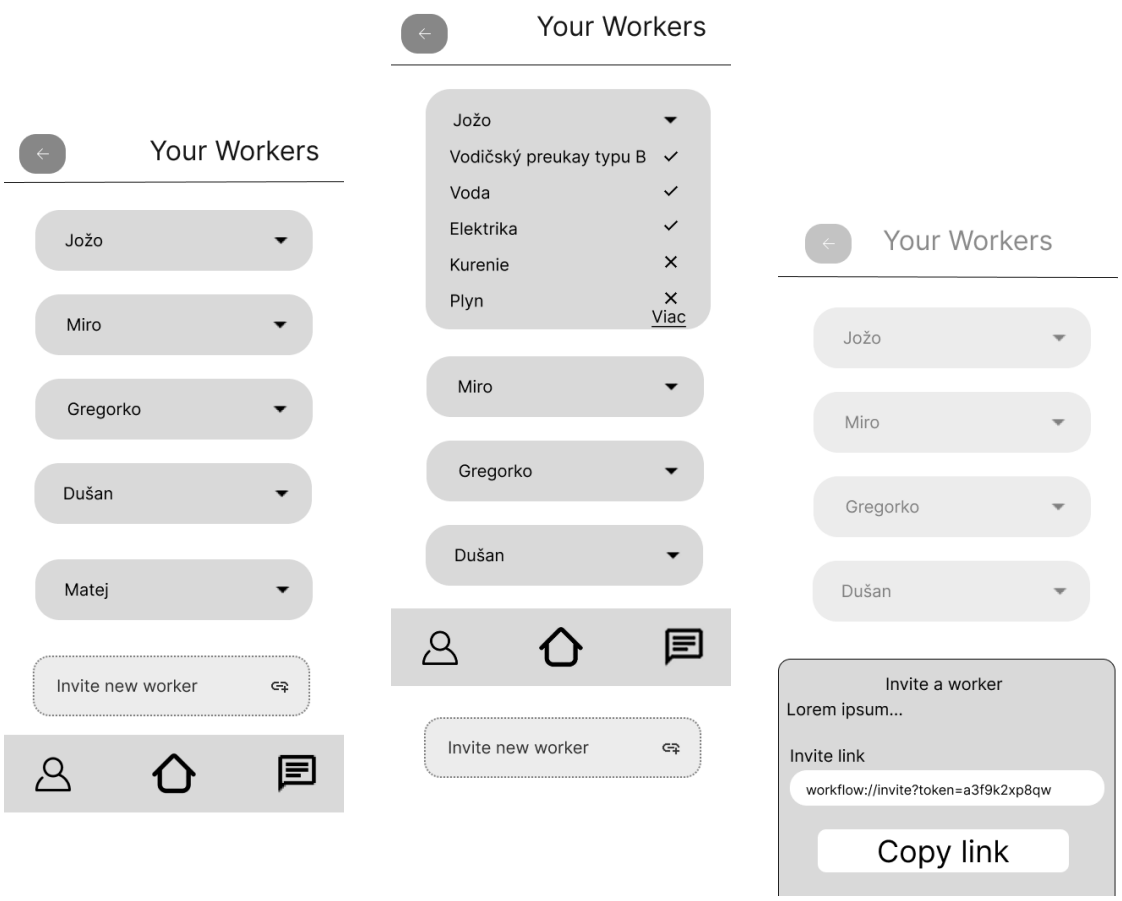
Task List, Task Detail and Task Creation



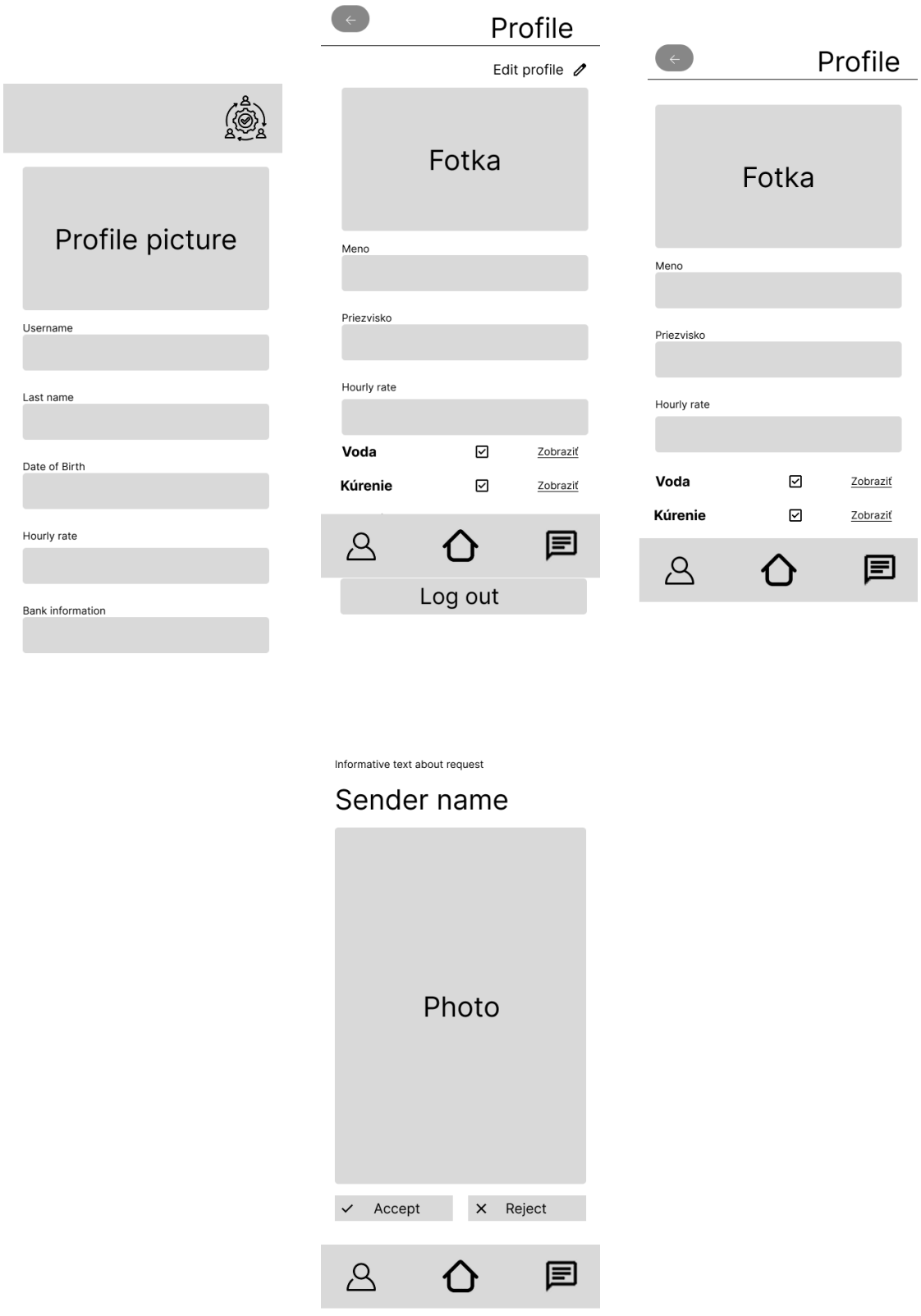
Assign Worker, Map and Chat



Worker List, Worker Detail and Invite Worker



Profiles and Certificates



Acceptance Tests (UAT)

Each test contains at least 7 steps, spans at least two screens, and represents a complete, self-contained scenario. Login and registration are not standalone UAT scenarios.

Positive Tests

UAT-01: Create a New Task

Preconditions: Manager is logged in. Active internet connection.
Postconditions: The new task is saved to the database and appears in the task list.

Step	Action
1	User is on the Task List screen.
2	Taps the + button (Create Task).
3	Create Task screen appears.
4	Fills in the title, description, and selects a deadline.
5	Taps on the map to set the task-site location.
6	Taps the **Save** button.
7	App navigates back to Task List; the new task appears at the top.
Result	PASS / FAIL

UAT-02: Assign a Worker to a Task

Preconditions: Manager is logged in. At least one Worker and one task exist.
Postconditions: Worker is assigned to the task and receives a push notification.

Step	Action
1	Manager opens Task List and taps an existing task.
2	Task Detail screen is displayed.
3	Taps **Assign Worker**.
4	A list of workers appears; Manager selects one.
5	Confirms the selection with the **Confirm** button.
6	The assigned worker's name appears in Task Detail.
7	Worker's device receives: *"You have been assigned a new task"*.
Result	PASS / FAIL

UAT-03: Worker Updates Task Status

Preconditions: Worker is logged in with a *Pending* task. Active internet.
Postconditions: Status changes to *In Progress*; Manager sees the update in real time.

Step	Action
1	Worker opens Task List (My Tasks).
2	Taps a task with status *Pending*.
3	Task Detail screen is displayed.
4	Taps the **Start Task** button.
5	Confirmation dialog appears; Worker taps **Confirm**.
6	Task status changes to *In Progress* with the appropriate badge.
7	On the Manager's device, the status updates in real time in Task List.
Result	PASS / FAIL

UAT-04: Upload a Photo to a Task

Preconditions: Worker is logged in with an assigned task. Active internet and camera permission granted.
Postconditions: Photo is uploaded and visible in Task Detail.

Step	Action
1	Worker opens Task Detail of an assigned task.
2	Scrolls to *Photo Documentation* and taps +.
3	Picker appears: *Camera* or *Gallery*; Worker selects *Camera*.
4	Camera opens; Worker takes a photo and confirms the shot.
5	App uploads the photo to the server.
6	A progress indicator is displayed during the upload.
7	After successful upload the photo appears in the task gallery.
Result	PASS / FAIL

UAT-05: Real-time Chat Between Manager and Worker

Preconditions: Both users are logged in with the chat of the same task open.
Postconditions: Messages appear on both devices in real time.

Step	Action
1	Manager opens Task Detail and taps **Chat**.
2	Worker opens the same task and taps **Chat**.
3	Manager types a message and taps **Send**.
4	Message appears instantly on the Worker's device without refreshing.
5	Worker replies and taps **Send**.
6	Message appears instantly on the Manager's device.
7	Both devices show the full history in correct chronological order.
Result	PASS / FAIL

UAT-06: Track Worker Location on the Map

Preconditions: Manager is on Map View. Worker is logged in with location sharing active.
Postconditions: Manager sees the Worker's live position on the map.

Step	Action
1	Manager opens Map View.
2	Task-site pins are visible on the map.
3	Worker moves physically (or simulator changes the location).
4	Worker opens the app — GPS transmission starts.
5	Worker marker appears on the Manager's map within 15 seconds.
6	After further movement, the marker shifts to the new position.
7	Manager taps the marker — Worker's name and status are shown.
Result	PASS / FAIL

UAT-07: Offline Task Status Update

Preconditions: Worker is logged in with an assigned task. No internet (airplane mode on).
Postconditions: After reconnection the status change is synced to the server.

Step	Action
1	Worker enables airplane mode.
2	Opens Task List — tasks are visible from local cache.
3	Opens Task Detail and taps **Mark as Done**.
4	App shows status *Done* with an offline sync icon (clock).
5	Worker disables airplane mode.
6	App detects connectivity and triggers background sync.
7	The clock icon disappears; task status is updated on the server.
Result	PASS / FAIL

Negative Tests

UAT-08: Worker Attempts to Create a Task

Preconditions: User with the Worker role is logged in.
Postconditions: App does not allow task creation; server returns 403.

Step	Action
1	Worker opens Task List.
2	The + (Create Task) button is not visible — hidden at UI level.
3	Attacker calls `POST /api/tasks` directly with the Worker's JWT.
4	Server verifies the user role.
5	Server returns `403 Forbidden`.
6	App displays: *"You do not have permission for this action."*.
7	No new task is created in the database.
Result	PASS / FAIL

UAT-09: Chat Without Internet Connection

Preconditions: User is logged in. No internet (airplane mode on).
Postconditions: Message is not sent; app informs the user of the error.

Step	Action
1	User enables airplane mode.
2	Opens Chat for a task.
3	Types a message and taps **Send**.
4	App attempts to establish a WebSocket connection.
5	Connection fails — banner appears: *"You are offline. Chat is unavailable."*.
6	Message is not sent and is not saved to local cache.
7	After internet is restored, the banner disappears and chat works normally.
Result	PASS / FAIL

UAT-10: Delete a Task with an Active Worker

Preconditions: Manager is logged in. An *In Progress* task with an assigned Worker exists.
Postconditions: After confirmation, the task is deleted and the Worker is notified.

Step	Action
1	Manager opens Task Detail of an *In Progress* task.
2	Taps **Delete Task**.
3	Warning dialog: *"This task has an assigned worker. Are you sure?"*.
4	Manager confirms the deletion.
5	Server deletes the task and its assignments from the database.
6	Manager is redirected to Task List — task is no longer visible.
7	Worker receives: *"A task was deleted by the Manager"* and it disappears from their list.
Result	PASS / FAIL